HT_MK1 Harness Tester

User Manual

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1 Introduction

1.1 Purpose

This manual explains how to operate the HT_MK1 harness tester using the GUI application: starting the software, connecting to the instrument, loading a netlist, running each test, and reading the results.

1.2 Audience

Operators running tests on the HT_MK1. No programming or electronics background is assumed.

1.3 Related Documents

- HT_MK1 GUI Functional Description: what every screen, section and button of the GUI does.
- HT_MK1 System Overview: what the instrument measures and how.
- HT_MK1 Code Walkthrough: for engineers maintaining the firmware or GUI software.
- HT_MK1 Flashing Procedure: for programming a unit's firmware.
- HT_MK1 Functional Test Procedure: for the formal acceptance test of a built unit.

2 System Overview

The HT_MK1 tests up to 256 wires in a harness. The GUI groups the tests into two modes and one survey tool:

Section	Test	Harness on	Checks
Mode 1	Continuity	Matrix connector (J-MTX)	Each wire is connected where it should be, and not touching any other wire.
Mode 1	Resistance	Matrix connector (J-MTX)	Each connected wire's resistance is within its allowed limit.
Mode 2	Insulation test	HV connector (J-HV)	Each wire holds the required insulation at 500 V DC, with no leakage.
Auto scan	Connection survey	Matrix connector (J-MTX)	Finds which pins are connected. It needs no netlist and is not a pass or fail test.

Table 1: Test sections

Mode 1 is run with the harness on the Matrix connector. Mode 2 is run with the harness moved to the HV connector. The GUI tracks which connector is in use and asks you to confirm each move. Mode 2 is normally unlocked only after Mode 1 passes. You can choose Continue anyway to run insulation without Mode 1; the skipped tests are recorded in the log.

3 Safety

3.1 High Voltage

WARNING: The insulation test applies up to 500 V DC to the harness under test. Never touch the harness, the HV connector, or the test fixture while a test is running or while the system is armed.

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While high voltage is present the GUI replaces its status bar with a red and yellow striped banner reading `HIGH VOLTAGE LIVE - J-HV ENERGIZED`.

3.2 Arming

Insulation testing needs an explicit confirmation before it runs. The GUI shows a checklist and a tick box; the Energize 500 V button stays disabled until you tick the box. The GUI then arms the instrument and starts the run. Arming on its own does not apply high voltage; running the test does. Before you tick the box, confirm the harness is on the HV connector and that no one is in contact with it.

3.3 Abort

To stop a test and remove high voltage, use one of these two controls. Both tell the instrument to stop, discharge any high voltage and open all relays:

- **Abort & discharge** in the high-voltage banner, shown whenever high voltage is live.
- **Emergency discharge** in the Safety panel of the Insulation screen. It is always available.

IMPORTANT: The round Stop button (and the Esc key) only resets the GUI. They do not send a stop command to the instrument. If you need to stop a test in progress, press Emergency discharge on the Insulation screen.

3.4 "Safe to Handle"

IMPORTANT: Wait for the green HV Safe indicator in the status bar before handling the harness. Do not assume it is safe simply because a test has finished. If the connection to the instrument is lost, treat the system as unsafe until it reconnects and shows HV Safe again. Right after connecting, the indicator reads HV state unknown until the instrument reports safe.

4 Getting Started

4.1 Launching the Application

Double-click the HT_MK1 GUI program. No separate installation is needed. The window opens maximized. (The Demo program has a built-in simulator and needs no instrument; nothing it shows is a measurement.)

4.2 Connecting to the Instrument

1. Connect the instrument to the PC with the USB cable and power it on.
2. In the status bar, open the port list and choose the instrument's serial port (the ST-LINK Virtual COM Port). Press the refresh button if it is not listed.
3. Press **Connect**. The link indicator turns green (`Link COM<n>`) and the state indicator shows Idle.
4. Enter the **DUT ID** and your name in **Operator**. They are written into every report made afterwards.

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NOTE: The DUT ID and Operator fields are shown only when the window is wide enough (more than 1600 logical pixels). If you cannot see them, widen the window.

4.3 Screen Layout

Area	Shows
Navigation rail (left)	Switches between screens: Run, Continuity and Resistance (Mode 1), Insulation (Mode 2), Auto scan, Netlist and Results.
Status bar (top)	Connection state, the loaded MTX and HV netlist names, which connector the harness is on, DUT ID and Operator, the HV Safe indicator and the state indicator.
Log bar (bottom)	The latest message. Press Expand to see all messages since start-up.

Table 2: Screen areas

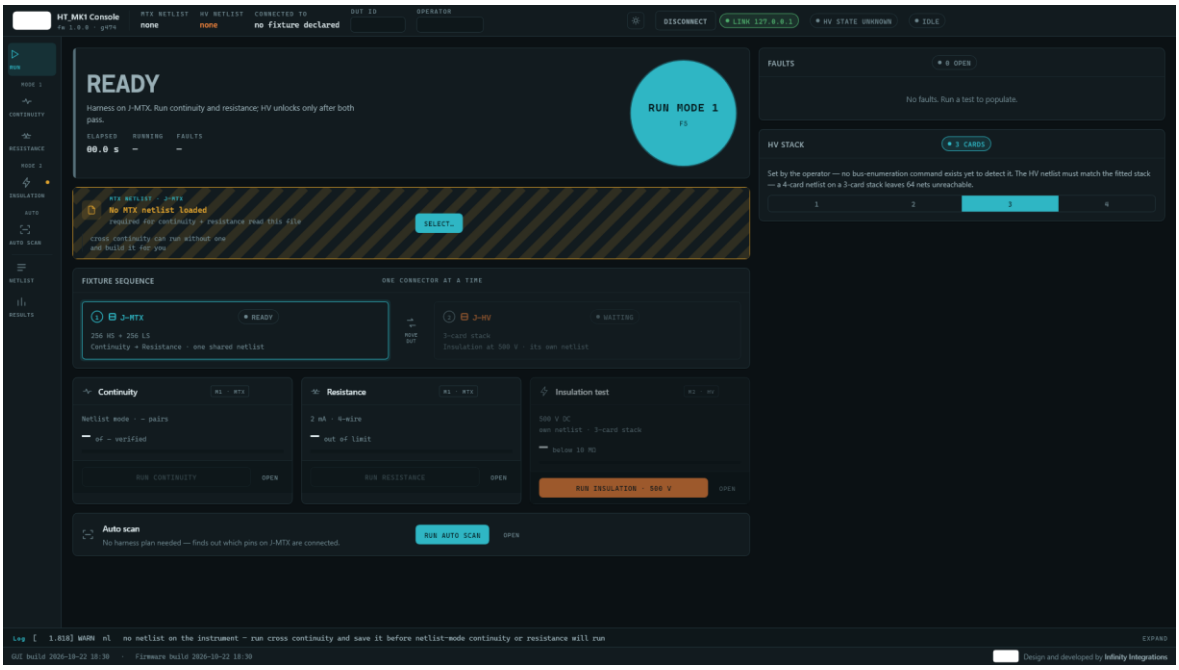


Figure 1: The Run screen when the GUI is first connected, before a netlist is loaded.

5 Loading a Netlist

A netlist is the list of wire connections a harness is expected to have. Continuity and resistance need the MTX netlist, and insulation needs the HV netlist. The instrument keeps the MTX pin pairs only until it is powered off, so load the netlist each session.

1. On the Run screen press **Select...** in the MTX netlist strip. (You can also use the Netlist screen.)
2. Press **Browse for a netlist file (.xlsx)...** and choose the file. Only Excel .xlsx files are accepted.
3. If the file contains connector information, the **Confirm the connector layout** dialog appears. Check that each connector has the right shape and name, then press **Apply**.
4. The strip now shows the file name and the number of connections. The log confirms `netlist uploaded to the instrument`.

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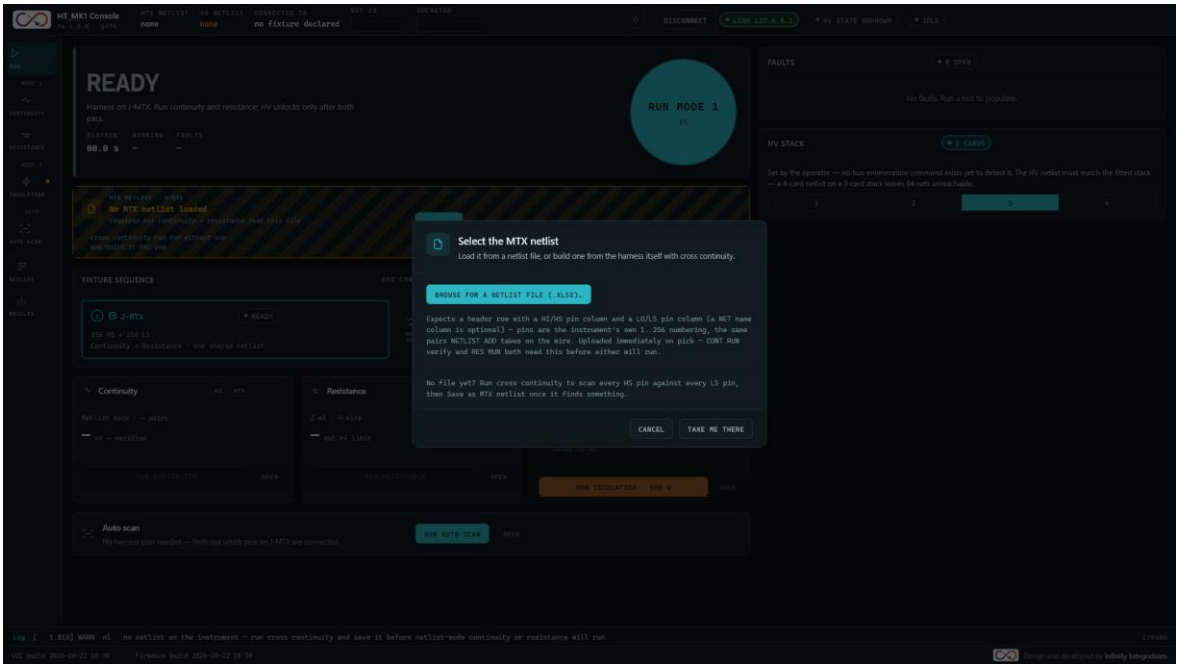


Figure 2: The Select the MTX netlist dialog.

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Figure 3: The Confirm the connector layout dialog.

The HV netlist is chosen when you start the insulation test (Section 6.2). It can be the same file as the MTX netlist. Before the first insulation run, also set the **HV stack** on the Run screen to the number of HV cards fitted in the instrument.

If you have no netlist file for a new harness type, use **Auto scan** (Section 6.3) to find the connections and save them as the MTX netlist.

The GUI reads an Excel file whose first sheet has a header row with a HI pin and a LO pin column, and optional columns for the net name, connectors, part numbers, pin labels and connector detail. The GUI Functional Description lists them all.

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6 Running a Test

6.1 Mode 1: Continuity and Resistance

1. Confirm the harness is plugged into the Matrix connector (J-MTX).
2. On the Run screen press **Run Mode 1** (or press F5). This starts the continuity test. Results appear live, pin by pin, and a wiring diagram and a results table update as each result arrives.
3. To use the Continuity screen instead, open it and press **Run continuity**. Choose **Netlist** mode to check the harness against the loaded netlist.

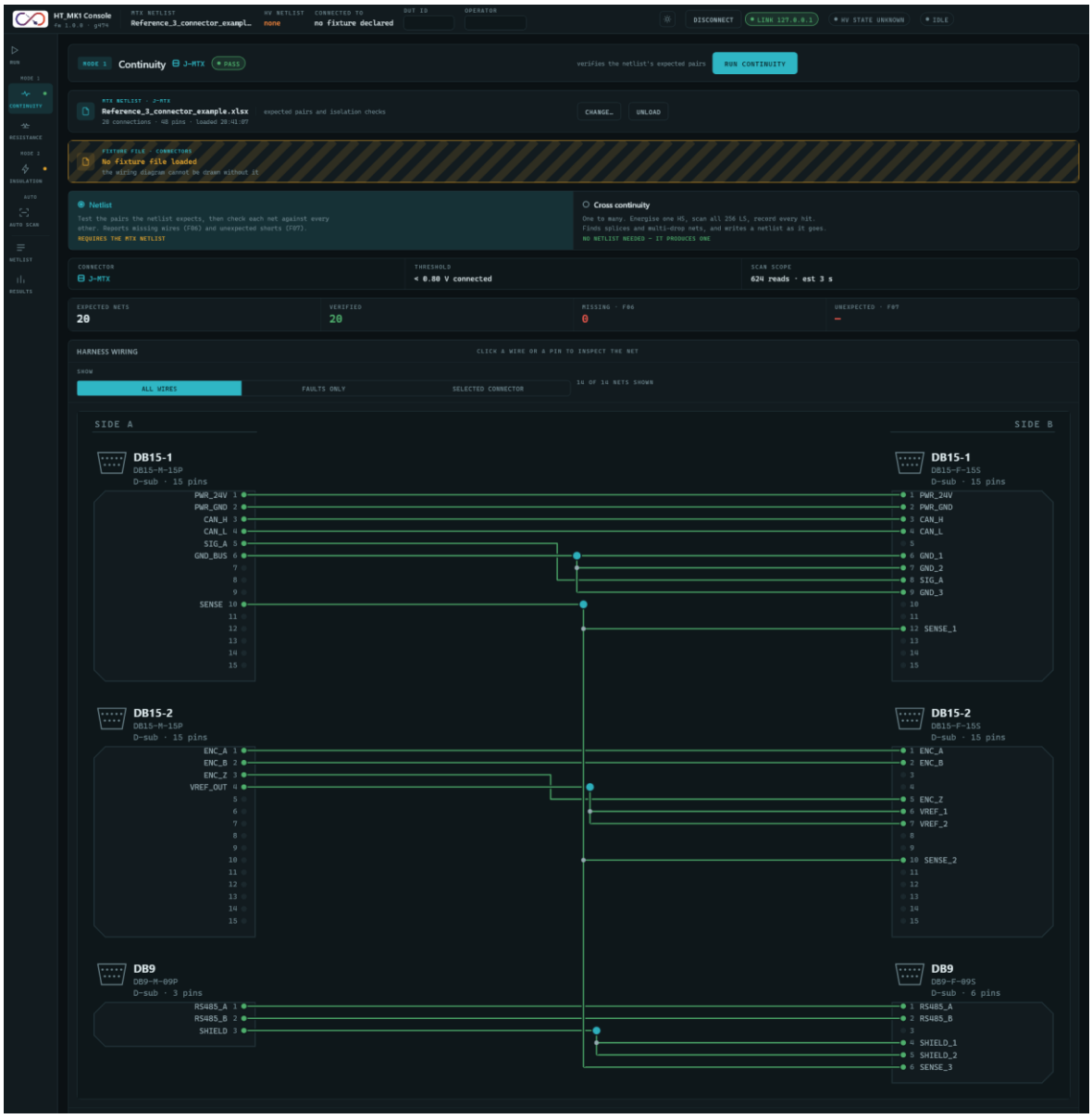


Figure 4: The Continuity screen after a passing run: result tiles and the wiring diagram.

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1. When continuity has passed, open the Resistance screen (or use the Run screen) and press **Run resistance**. Each wire's result appears live, and a table ranks the wires by how close they are to the limit.

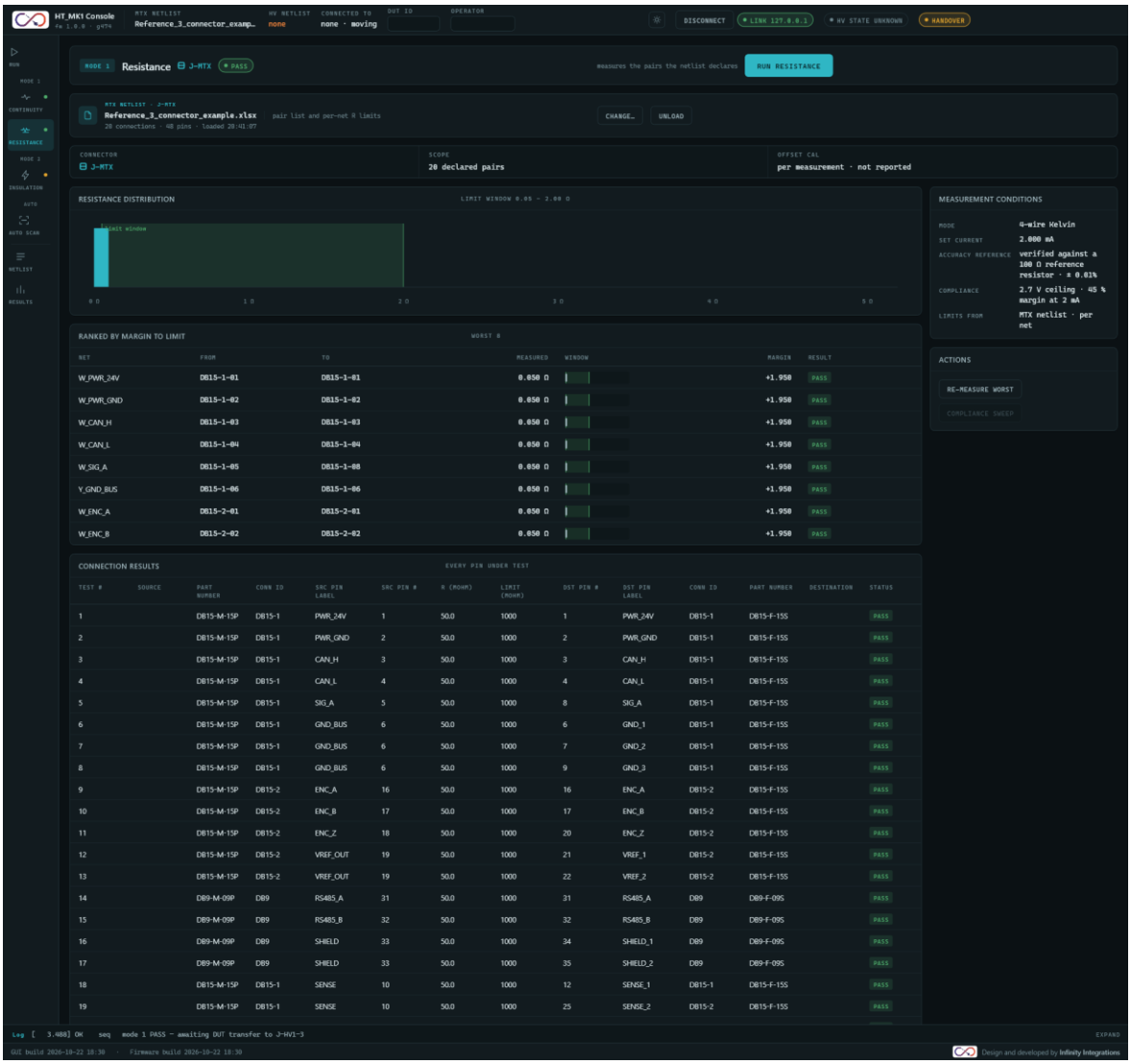


Figure 5: The Resistance screen after a passing run, ranked by margin to the limit.

When both tests pass, the Run screen shows **MOVE DUT** and asks you to move the harness to the HV connector (Section 6.2). If either test fails, the screen shows **FAIL**, the harness stays where it is and high voltage is not attempted.

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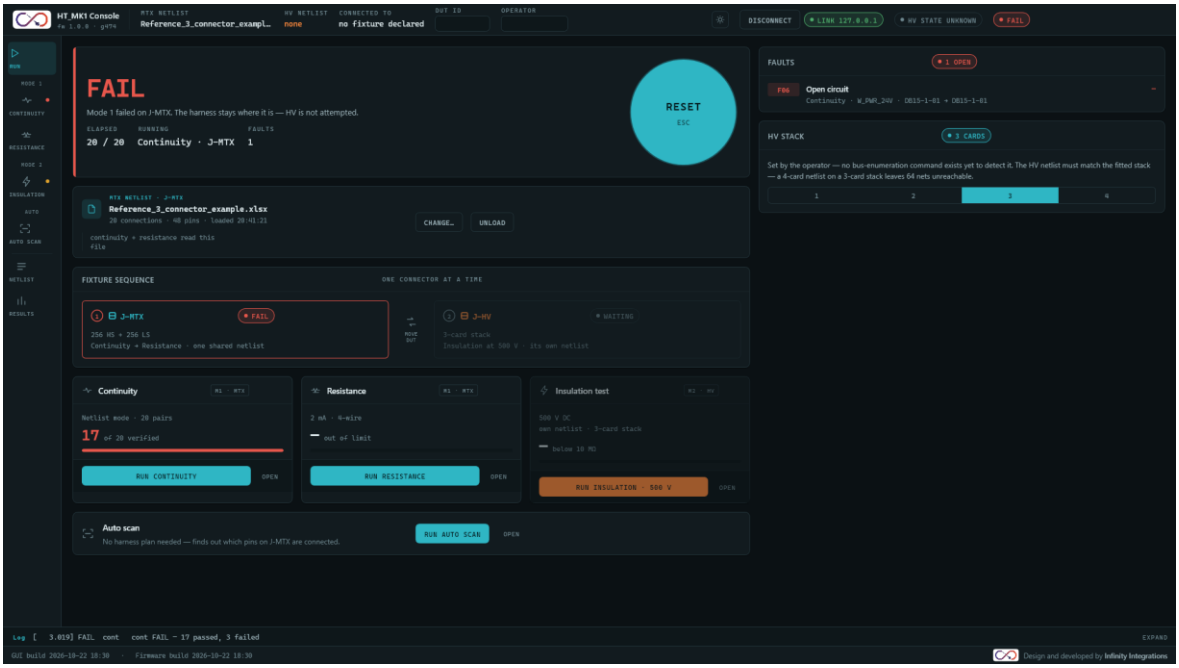


Figure 6: The Run screen after a failed continuity run: the verdict shows FAIL and the Faults panel lists the open nets.

6.2 Mode 2: Insulation

WARNING: Read Section 3 before running this test.

Once continuity and resistance both pass, the Run screen asks for the fixture handover:

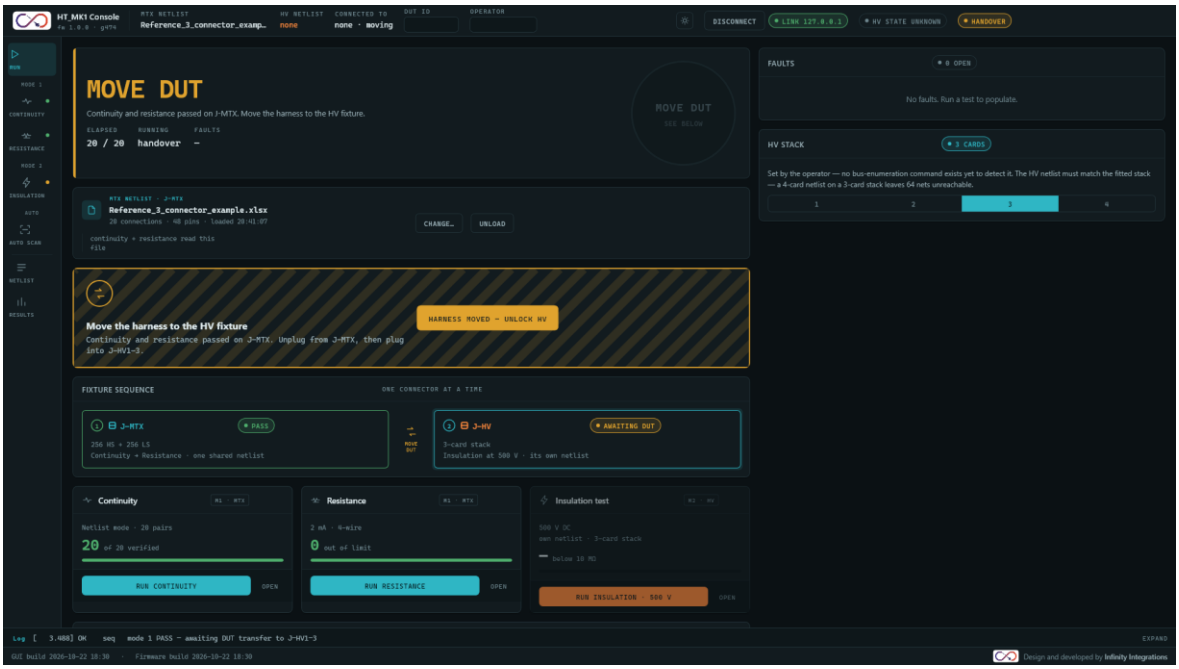


Figure 7: The handover prompt shown once Mode 1 passes.

1. Move the harness from J-MTX to the HV connector (J-HV).
2. Press **Harness moved - unlock HV**. The verdict changes to ARMED and the Insulation test is unlocked.

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3. Check that the **HV stack** setting on the Run screen matches the HV cards fitted.

4. Press **Run insulation** (on the Run screen or the Insulation screen). If no HV netlist is loaded, choose one when asked.

5. A confirmation dialog lists the checks (continuity, resistance, fixture, netlist match, interlock). Read every row. Tick the confirmation box. The **Energize 500 V** button stays disabled until you do.

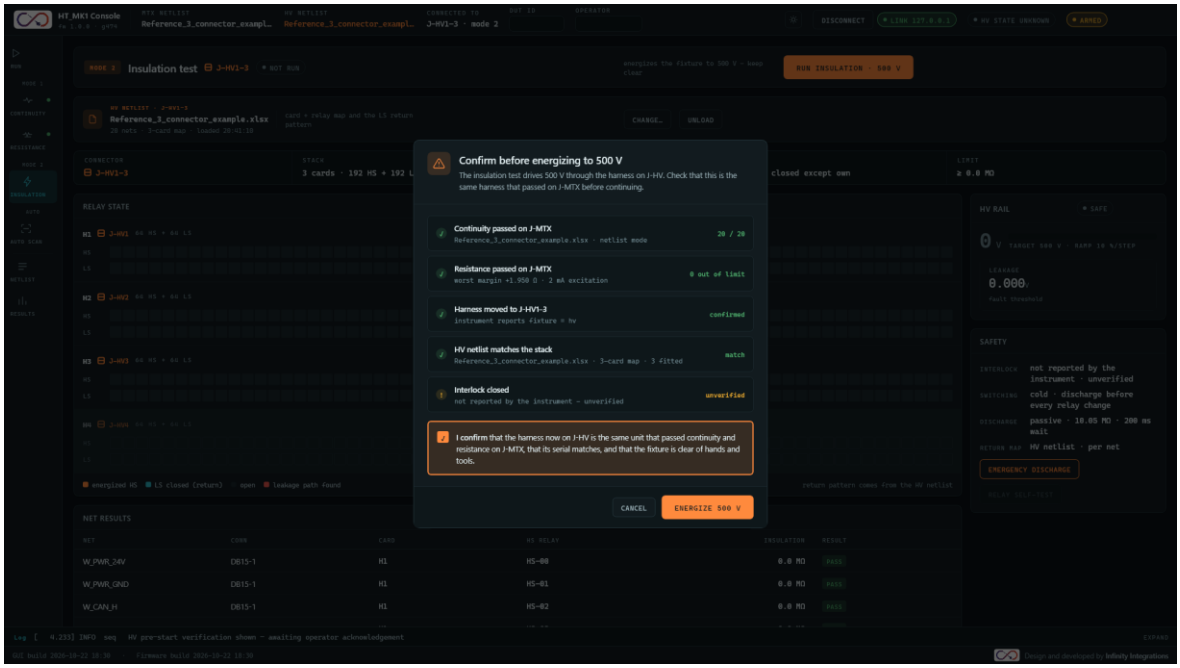


Figure 8: The pre-energize confirmation. Energize 500 V is enabled only after the box is ticked.

1. Press **Energize 500 V**. The instrument applies high voltage to each wire in turn and the GUI shows the results live. While voltage is present the status bar is replaced by the high-voltage banner.

2. Wait for the green **HV Safe** indicator before handling the harness.

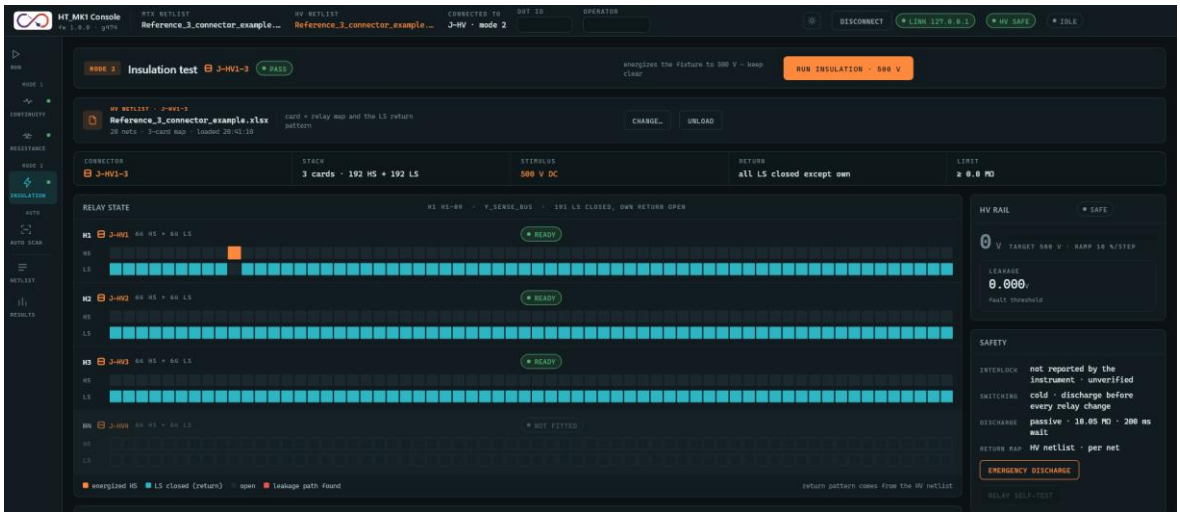


Figure 9: The Insulation screen after a run: relay state and HV rail.

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Continue anyway. If you press Run insulation before Mode 1 has passed, the GUI asks `Run continuity and resistance first?` You can go to Mode 1, cancel, or choose **Continue anyway** to run insulation without any pre-test. Continue anyway tells the instrument the harness is on J-HV, so make sure it is plugged in there. The skipped tests are recorded in the log and shown as `skipped` in the confirmation dialog.

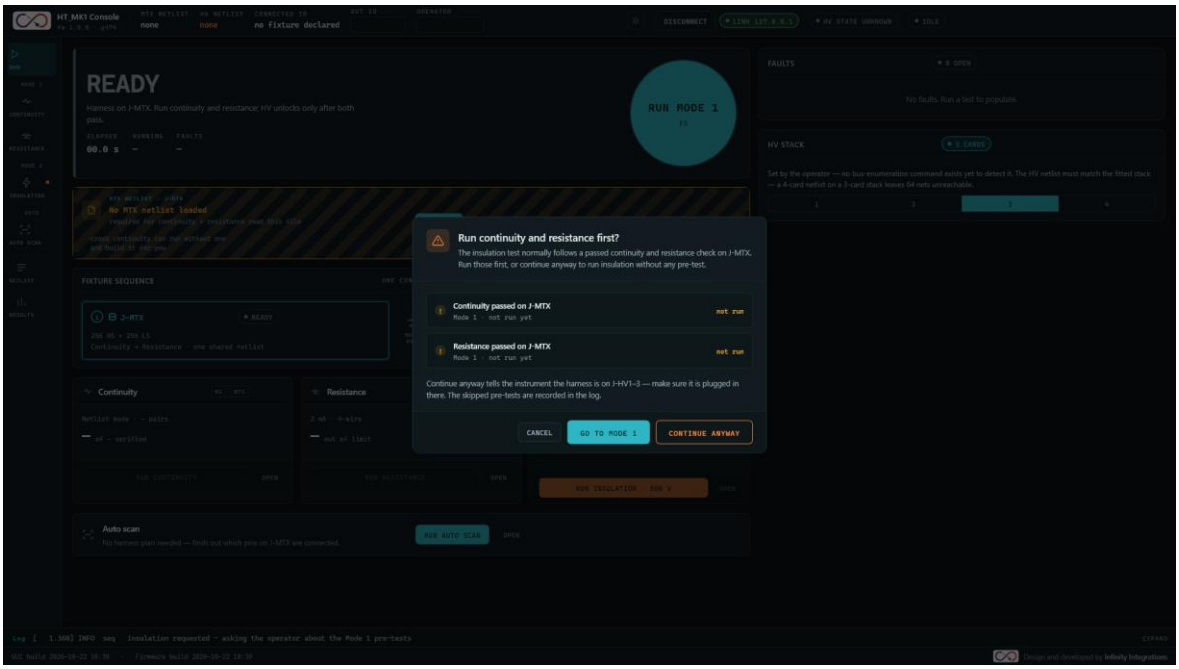


Figure 10: The prompt shown when insulation is requested before Mode 1 has passed.

6.3 Auto Scan

Auto scan finds the connections that exist on a harness without a netlist. Use it for a new or unknown harness.

1. Plug the harness into the Matrix connector (J-MTX).
2. Open the **Auto scan** screen and press **Run auto scan**.
3. The connections found appear in the table. When the scan ends, the tiles show the number of connections, pins and multi-drop nets.
4. To use the result as a netlist, press **Save as MTX netlist**. It is then loaded as the MTX netlist.

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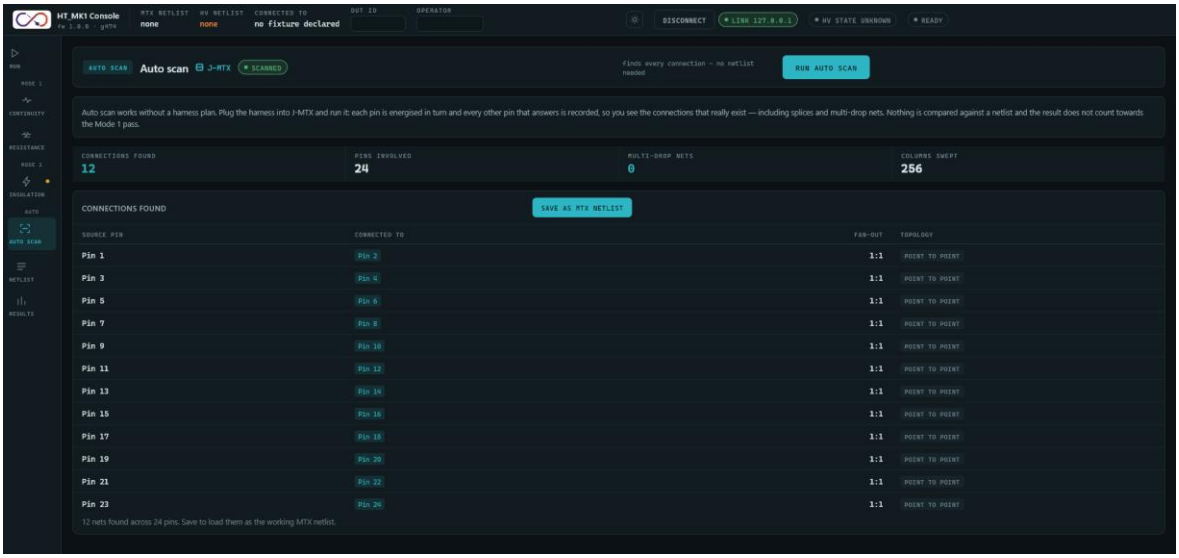


Figure 11: The Auto scan screen after a scan.

NOTE: Auto scan is a survey. It does not count toward the Mode 1 pass and cannot unlock insulation.

6.4 Reading Results

Each wire shows a pass or fail result as it arrives. The Run screen shows a verdict, the three test cards show running counts, and the Faults panel lists any failures. Pressing a fault takes you to the screen that shows it. A small dot on the navigation rail shows each test's last result: green for pass, red for fail.

Fault codes:

Code	Meaning
F06	Open circuit: an expected connection was not found.
F07	Unexpected connection (short) between wires that should not be joined.
F08	Resistance outside the limit.
F04	Insulation below the limit.

Table 3: Fault codes

7 Viewing Results and History

7.1 Run History

The Results screen lists past test runs, most recent first, stored on the PC running the GUI, with a recent pass-rate summary. History is kept when the GUI is closed.

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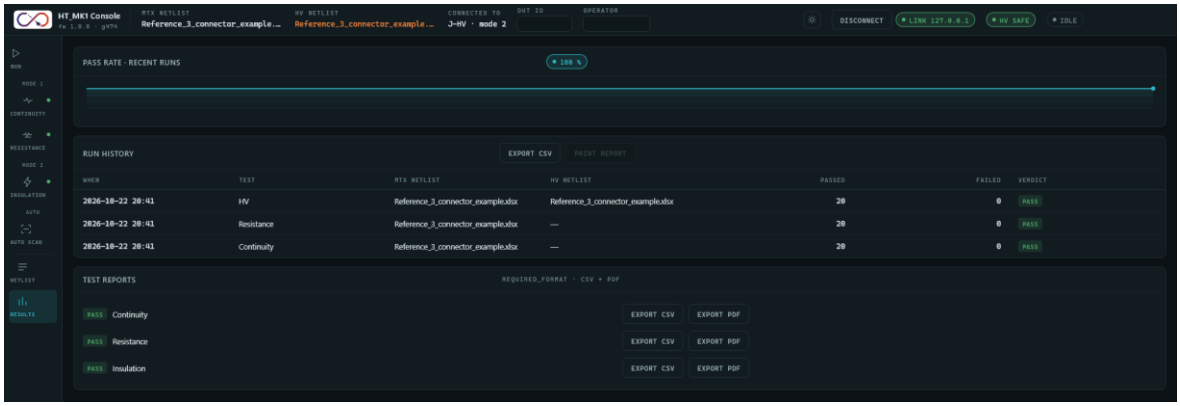


Figure 12: The Results screen: pass rate, run history and per-test reports.

7.2 Exporting a Report

1. Go to the **Results** screen.
2. In **Test reports**, find the test (Continuity, Resistance or Insulation).
3. Press **Export CSV** or **Export PDF** and choose where to save the file.

NOTE: A report is kept in memory for the most recent completed run of each test. It is lost when the GUI is closed, so export your reports before closing. Enter the DUT ID and Operator before the run so they appear in the report. **Export CSV** next to Run history saves the run history.

8 Troubleshooting

8.1 Common Messages

The GUI reports problems as lines in the log bar at the bottom of the window, not as pop-up windows. If a button seems to do nothing, press **Expand** on the log bar and read the newest lines.

Message (in the log)	Meaning	What to do
not connected	You pressed Run with no link to the instrument.	Choose the port and press Connect.
refused - EBUSY	Another test is still running.	Wait for it to finish, or press Emergency discharge.
refused - EFIXTURE	The harness is on the wrong connector for the requested test.	Move the harness to the correct connector and confirm it in the GUI.
refused - ENOTARMED	Insulation was requested without arming.	Start insulation from the GUI as in Section 6.2.
refused - ERANGE	A value was outside the allowed range.	Re-enter a valid value.
refused - EHW	The instrument failed to initialize one of its cards at power-on.	Power-cycle the instrument and check the card connections. If it persists, contact support.
could not read <file>	The netlist file could not be read.	Check it is an .xlsx file with a HI and a LO pin column and pin numbers from 1 to 256.
netlist upload refused	The instrument did not accept the netlist.	Check the netlist and try again.
no netlist on the instrument	No netlist is held by the instrument.	Load a netlist, or run Auto scan and save it.

Table 4: Common messages

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8.2 If the Connection Is Lost

The status bar shows **No link**, **Link lost** and **State unknown**, and the log shows `link lost`. The GUI does not reconnect by itself. Check the cable, then choose the port and press **Connect** again. Treat the system as not safe to handle until **HV Safe** is shown again.

8.3 Faults

If the instrument latches a fault, the state indicator turns red and reads **Fault**, and a **Clear Fault** button appears in the status bar. The instrument refuses every run until the fault is cleared. Remove the cause (for example reseat a card), then press **Clear Fault**. The state returns to Idle. A fault is never cleared automatically.

9 Glossary

Term	Meaning
Netlist	The list of expected wire connections for the harness under test.
MTX netlist	The netlist for continuity and resistance on the Matrix connector (J-MTX).
HV netlist	The netlist for insulation on the HV connector (J-HV). It follows the number of HV cards fitted.
Mode 1	Continuity and resistance, with the harness on J-MTX.
Mode 2	The insulation test, with the harness on J-HV.
Auto scan	A survey that finds the connections on a harness without a netlist.
Fixture	Which physical connector the harness is on: Matrix (J-MTX) or HV (J-HV).
Arm	An explicit step required before high voltage can be applied. The GUI does it when you energize.
Verdict	The pass or fail result for one wire or for a whole test.
Leakage	The small unwanted current that flows through imperfect insulation.

Table 5: Glossary

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